

RESEARCH ARTICLE

The role of the power law exponent in wind energy assessment: A global analysis

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Summary

The power law is most often applied to extrapolate the near-surface wind speed to the wind turbine hub height. Due to variations of the meteorological conditions, the power law exponent varies over time. Usually, no long-term wind speed measurements from multiple heights are available which would allow time-dependent and spatially explicit power law exponent estimations. Instead, often the mean of the power law exponent or a constant value of 0.14 is assumed. The goal of this study was to quantify the error in wind potential assessments resulting from applying the mean of the power law exponent or a value of 0.14. The data base for this study are the hourly wind speed time series at 10 and 100 m above ground available for the period 2007 to 2018 from the ERA5 reanalysis project at a global $0.25^\circ \times 0.25^\circ$ grid. The errors in the estimation of the wind power density and the capacity factor were calculated. It was found that, onshore, the global median of the absolute percentage error related to the wind power density using the mean of the power law exponent is 7.5%. Assuming a constant value of 0.14, the power law is less accurate (absolute percentage error: 37.1%). For the estimation of the capacity factor the absolute percentage errors are 5.5% and 36.9%. Based on the results of this study, the use of time-dependent and spatially explicit power law exponents is suggested. In the absence of long-term wind speed measurements from multiple heights, the results provide a comprehensive global overview of the errors to be expected from using the mean of the power law exponent or assuming a value of 0.14. In many regions where the wind resource is abundant, using the mean of the power law exponent only leads to minor errors in capacity factor estimation. There, the assessment of wind resources with small errors is possible, even in the absence of long-term wind speed measurements at different heights.

KEYWORDS

method error assessment, seasonal cycle, Wakeby distribution, wind resource, wind speed extrapolation

ABBREVIATIONS: DJF, December, January, February; JJA, June, July, August; EV, variant of the power law exponent; GloWiSMo, Global Wind Speed Model; LIDAR, Light Detection and Ranging; MAM, March, April, May; SON, September, October, November; Wak, five-parameter Wakeby distribution.

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1 | INTRODUCTION

The ongoing global warming requires a rapid reduction in global greenhouse gas emissions. An important pillar for mitigating climate change is the conversion of the power supply based on conventional to renewable energies. The technical wind potential is sufficient to cover global electricity consumption by wind energy.^{1–3} Thus, one of the most effective renewable energies for the mitigation of climate change is wind energy.⁴

The need for climate change mitigation and the technical progress of wind turbine technology has led to an increasing wind energy expansion in recent years. In 2019, the globally installed wind capacity amounted to 651 GW.⁵ Since 2014, the annual expansion rate of wind capacity exceeded 50 GW.⁵

However, the current wind energy expansion rate is still not sufficient to meet the goals set by the Paris Climate Agreement aiming at lowering greenhouse gas emissions. The comprehensive exploitation of the globally available technical wind potential requires the installation of much more wind turbines.³ The installation of wind turbines cannot always be as smooth and efficient as it should be, as their installation is highly regulated in some parts of the world. Barriers to the installation of wind turbines include the lack of detailed knowledge on wind resources,⁶ geographical restrictions,⁷ availability of wind turbine technology,⁸ and socio-economic conditions.⁹

The basis for wind potential assessments is detailed knowledge of wind resources. Globally, wind resources are very unevenly distributed. The areas where the wind resources are abundant include the southern tip of South America, the Midwest of the USA, Canada, Russia, and many deserts.³ Besides, the wind resources are abundant off many coasts and inshore.¹⁰

The dynamics of wind resources are primarily driven by large-scale circulation systems.¹¹ On the local scale, surface roughness and orography then modify the large-scale wind resource patterns.¹² As a result, exposed mountain ranges and areas where the surface roughness is low, often provide sufficient wind resource for wind turbine installations.¹³

Besides the great spatial variability, an inherent feature of wind resources is their temporal variability.¹⁴ It extends over several time scales, ranging from sub-hourly¹⁵ to multi-decadal scales.¹⁶ Considering the temporal variability of wind resources in wind potential assessments is crucial because it is a key factor regarding their capability to provide a constant power supply.¹⁷

To capture the complex spatiotemporal pattern of the wind resource, detailed modeling, and measuring the wind speed (U) is required. Routine measurements of the

near-surface wind speed (U_g) have been carried out by national weather services for several decades.¹⁸ Usually, U_g measurements are made at a height of 10 m above ground level. However, the wind energy is commonly exploited in the wind turbine hub height (h_{hub}) range from 80 to 140 m above ground level. Thus, an important step in every wind potential assessment is the extrapolation of U_g to h_{hub} .

The most common approach to extrapolate U_g to h_{hub} is the power law¹⁹:

$$U_{hub} = U_g \left(\frac{h_{hub}}{h_g} \right)^E, \quad (1)$$

where h_g is the measurement height of U_g , E is the power law exponent, and U_{hub} is the wind speed in h_{hub} . The value assigned to E determines the increase of U with increasing height above the ground. Just as U , E is characterized by spatiotemporal variations which are driven by thermally and mechanically induced mixing, wind speed, air temperature conditions, elevation, time of day, season, and orography.²⁰ Usually, under a stably stratified atmosphere and/or high surface roughness, E values are high.²¹ In contrast, under unstable atmospheric conditions and/or when the surface roughness is low, E values are low.²² At greater values of E , U increases more with increasing height.

Since U_{hub} is very sensitive to E , the values of E impact the accuracy of wind energy assessments.²³ To estimate E values at a specific site, wind speed measurements in at least two different heights are required. Moreover, these measurements should be carried out over a period as long as possible to account for the temporal variability of E .²⁴ For instance, using a Light Detection and Ranging (LIDAR) system, long-term U observations were obtained in Singapore²⁵ and Ohio (USA)²⁶ to calculate spatiotemporal explicit E values. Another approach to calculate spatiotemporal explicit E values is to use U measurements from a tower equipped with U measurement sensors.²⁷ It is usually not possible to realize such measurements, due to logistical and financial reasons.

There are different approaches to substitute missing empirical values of E . The simplest, but frequently applied approach is the assumption that E is constant at the value of 0.14 ($E014$).²⁸ The advantage of using $E014$ is that neither measurements nor knowledge of the local influences on E is required. However, using $E014$ ignores both the temporal and spatial variability of E .

Another widely used approach to substitute missing empirical values of E is to implement spatially explicit but temporally constant mean values of E (EM).²³

Usually, EM values can be assigned to a site based on the local surroundings.²⁹ The use of EM neglects the temporal variability of E .

Previous studies reveal that applying the power law is fundamental in wind potential assessments and using EM or $E014$ as the power law exponent is widespread. Thus, quantifying the errors in the application of EM and $E014$ is of utmost importance to judge the accuracy of wind potential estimates. Unfortunately, such investigations are rare and have so far only been carried out for individual sites.²³ To obtain a generalizable error evaluation, it is essential to conduct investigations on a global scale. Besides, the regionally influenced temporal variability of E must be considered as a potential cause of EM and $E014$ errors.

Thus, the goals of this study are (a) the evaluation of the spatiotemporal variability of E and (b) the quantification of the error in the application of EM and $E014$ for wind resource and technical wind potential assessments.

2 | MATERIAL AND METHODS

The methodology to assess the influence of E on the accuracy of wind potential assessments includes the following steps (Figure 1): (a) Obtaining wind speed time series in 100 m (U_{100}) and 10 m (U_{10}) on a global grid, (b) calculation of E from U_{100} and U_{10} , (c) calculation of EM , (d) extrapolation of U_{10} to $h_{hub} = 100$ m by assuming variants of E yielding U_{100} , U_{EM} , and U_{E014} , (e) investigation of the seasonal variability and inter-

annual variability of EM , (f) calculation of the mean wind power density (WPD) using U_E (WPD_E), U_{EM} (WPD_{EM}), and U_{E014} (WPD_{E014}), (g) estimation of the probability density functions $f()$ of U_E ($f(U_E)$), U_{EM} ($f(U_{EM})$), and U_{E014} ($f(U_{E014})$), and (h) calculation of the capacity factors (cf) using $f(U_E)$, $f(U_{EM})$, and $f(U_{E014})$ yielding cf_E , cf_{EM} , and cf_{E014} by applying an exemplary wind turbine power curve.

2.1 | Wind speed data

Time series of hourly wind vector components (u (zonal), v (meridional)) in 10 m (u_{10} , v_{10}) and 100 m (u_{100} , v_{100}) in the period January 1, 2007 to December 31, 2018 were obtained from the ERA5 reanalysis project³⁰ for all onshore areas and offshore regions with sea depth < 200 m.

The horizontal resolution of u and v is $0.25^\circ \times 0.25^\circ$. The wind speed in 10 and 100 m was calculated by:

$$U_{he} = \sqrt{u_{he}^2 + v_{he}^2}. \quad (2)$$

with $he = \{10 \text{ m}, 100 \text{ m}\}$ being the height above ground level.

2.2 | The power law

Hourly values of E available at the grid cells were computed from U_{10} and U_{100} by Reference 19:

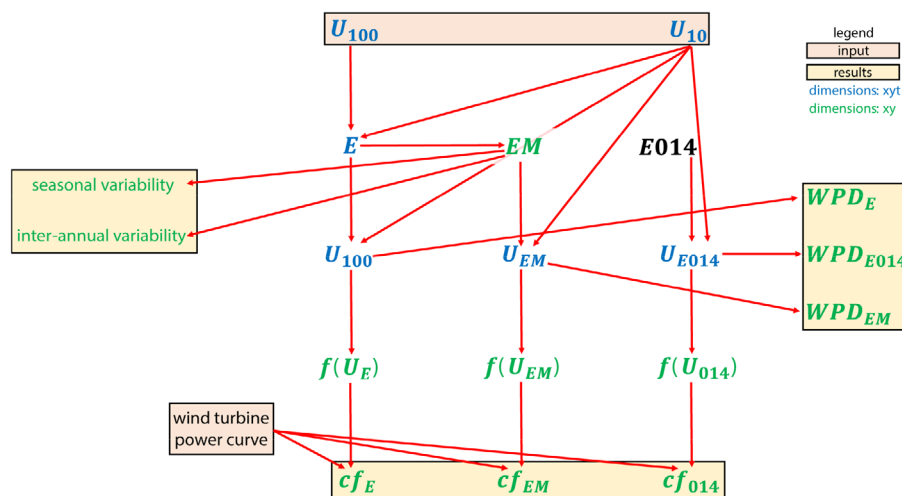


FIGURE 1 Methodology applied to assess the influence of the power law exponent (E) on the accuracy of wind potential assessments with U_{100} being the wind speed in 100 m, U_{10} being the wind speed in 10 m, EM being the mean of E , $E014$ being $E = 0.14$, WPD being the wind power density, $f()$ being the probability density function and cf being the capacity factor. The subscripts denote the E variants ($EV = \{E, EM, E014\}$). Gridded data available as time series (dimensions: xyt) is labeled in blue and time-independent gridded data (dimensions: xy) is labeled in green [Colour figure can be viewed at wileyonlinelibrary.com]

$$E = \frac{\ln\left(\frac{U_{100}}{U_{10}}\right)}{\ln\left(\frac{100m}{10m}\right)}. \quad (3)$$

From the time series of E , EM over the entire investigation period was calculated. The dimensions of $E014$, EM , and E differ: E varies at the grid cells and in time (t), EM varies only at the grid cells, and $E014$ is a constant. Hourly extrapolated wind speed in 100 m was calculated based on the variants of E by applying the power law¹⁹:

$$U_E = U_{10} \left(\frac{100m}{10m} \right)^E, \quad (4)$$

$$U_{EM} = U_{10} \left(\frac{100m}{10m} \right)^{EM}, \quad (5)$$

$$U_{E014} = U_{10} \left(\frac{100m}{10m} \right)^{E014}. \quad (6)$$

2.3 | Analysis of the temporal variability of the power law exponent

The temporal variability of E was assessed at seasonal and inter-annual scales. The seasonal behavior of E was evaluated for (a) December, January, February (DJF), (b) March, April, May (MAM), (c) June, July, August (JJA), and (4) September, October, November (SON). EM was calculated for each season yielding EM_{DJF} , EM_{MAM} , EM_{JJA} , and EM_{SON} .

The investigation of the inter-annual variability of E is based on the annual mean values of E for the entire investigation period. Both, the seasonal and inter-annual variability of E were quantified by the standard deviation (sd) and the range (r).

2.4 | Influence of the power law exponent on the wind resource and technical potential

To estimate the influence of the variants of E on wind resource assessments, WPD_E , WPD_{EM} , and WPD_{E014} were computed by Reference 20:

$$WPD_{EV} = 0.5 \cdot U_{EV}^3 \cdot AD \quad (7)$$

with AD being the air density ($AD = 1.225 \text{ kg/m}^3$).

The technical wind potential was quantified by cf . Before the calculation of the cf variants, the time series of U were fitted to the five-parameter Wakeby (Wak)³¹

distribution by the L-moment method.³² Wak is a very flexible distribution able to reproduce a greater variety of empirical distributions compared to other two- or three-parameter distributions.³¹ Thus, Wak is often designated as grandparent distribution.

In a recent review, Wak was found to provide a reliable fitting accuracy for U regimes around the world.³³ Wak consists of a location parameter (ϵ), two scale parameters (α , γ), and two shape parameters (β , δ). The probability density function of Wak is defined as:

$$f(U; \alpha, \beta, \gamma, \delta, \epsilon) = \left\{ \alpha [1 - F(U)]^{\beta-1} + \gamma [1 - F(U)]^{-\delta-1} \right\}^{-1}. \quad (8)$$

Most often Wak is defined by its quantile function:

$$F^{-1}(F; \alpha, \beta, \gamma, \delta, \epsilon) = \epsilon + \frac{\alpha}{\beta} \left[1 - (1 - F)^{\beta} \right] - \frac{\gamma}{\delta} \left[1 - (1 - F)^{-\delta} \right]. \quad (9)$$

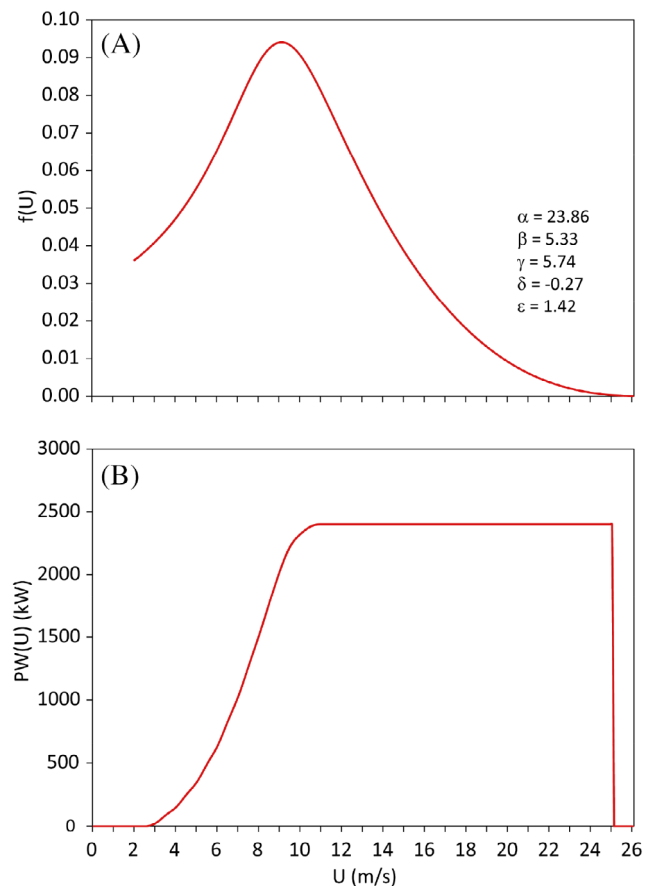


FIGURE 2 A, Schematic representation of a Wakeby probability density function $f(U)$ and B, the power curve ($PW(U)$) of the Nordex N117/2400 wind turbine [Colour figure can be viewed at wileyonlinelibrary.com]

For this study, the power curve ($PW(U)$) of the Nordex N117/2400 wind turbine was selected³⁴ (Figure 2B). This wind turbine is one of the most frequently installed wind turbines in Germany in recent years. It has a hub height of 100 m, a rated power (PR) of 2400 kW, and a rotor diameter of 117 m.

The calculation of the mean power output (PWM) was carried out using $PW(U)$ and $f(U_{EV})$:

$$PWM_{EV} = \int_0^{\infty} PW(U) f(U_{EV}) dU. \quad (10)$$

After PWM_{EV} estimation, cf was calculated by:

$$cf_{EV} = PWM_{EV} / PR. \quad (11)$$

The power law accuracy applying EM and $E014$ was evaluated using four quality measures. The quality measures are based on the comparison between the mean of the wind potential variables $WPV = \{cf, WPD\}$ in the investigation period that were calculated from hourly E (WPV_E) and from the variants ($EV = \{EM, E014\}$) of E (WPV_{EV}).

The first quality measure is the error (ER):

$$ER_{WPV} = WPV_{EV} - WPV_E. \quad (12)$$

The second quality measure is the absolute error (AER):

$$AER_{WPV} = |WPV_{EV} - WPV_E|. \quad (13)$$

The third quality measure is the percentage error (PER):

$$PER_{WPV} = \frac{WPV_{EV} - WPV_E}{WPV_E} \cdot 100. \quad (14)$$

The fourth quality measure is the absolute percentage error ($APER$):

$$APER_{WPV} = \left| \frac{WPV_{EV} - WPV_E}{WPV_E} \right| \cdot 100. \quad (15)$$

3 | RESULTS AND DISCUSSION

3.1 | Mean power law exponent

The global distribution of EM is displayed in Figure 3. The results reveal that the land-sea distribution is the most important factor influencing EM . Frequently, the

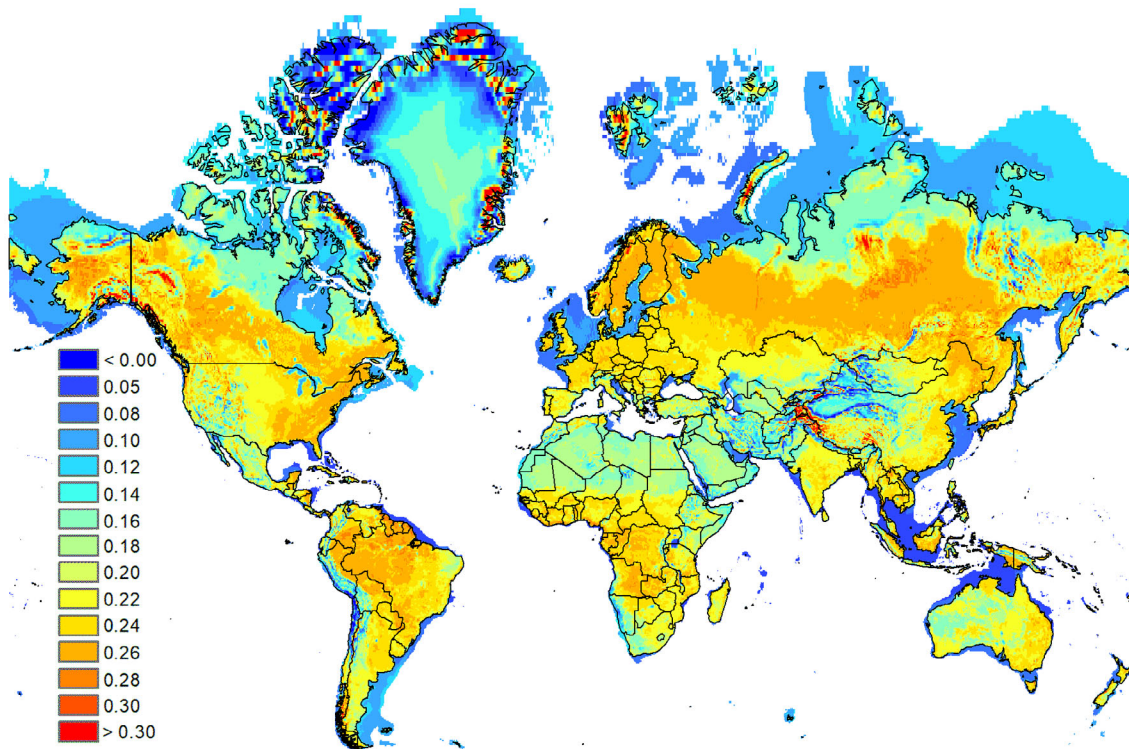


FIGURE 3 Global pattern of the mean power law exponent (EM). The legend values indicate the highest class values [Colour figure can be viewed at wileyonlinelibrary.com]

TABLE 1 Country-related median of the mean power law exponent ($\tilde{EM}$) of the countries with the largest populations

Country	$\tilde{EM}$	country	$\tilde{EM}$	country	$\tilde{EM}$
China	0.203	Canada	0.209	Chad	0.179
India	0.210	Afghanistan	0.147	Rwanda	0.184
USA	0.216	Morocco	0.153	Belgium	0.225
Indonesia	0.208	Malaysia	0.211	Burundi	0.213
Brazil	0.238	Venezuela	0.222	Tunisia	0.166
Pakistan	0.157	Peru	0.180	Cuba	0.204
Nigeria	0.226	Uzbekistan	0.166	Bolivia	0.233
Bangladesh	0.211	Nepal	0.133	Benin	0.232
Russia	0.231	Angola	0.237	Portugal	0.211
Japan	0.217	Saudi Arabia	0.167	Greece	0.172
Mexico	0.178	Yemen	0.151	Dominican Republic	0.208
Ethiopia	0.183	Ghana	0.219	Czechia	0.235
Philippines	0.173	Mozambique	0.218	Haiti	0.141
Egypt	0.169	North Korea	0.214	Jordan	0.157
Vietnam	0.220	Madagascar	0.210	Azerbaijan	0.165
Congo	0.236	Cameroon	0.227	Sweden	0.250
Iran	0.145	Ivory Coast	0.235	Hungary	0.229
Turkey	0.181	Taiwan	0.155	Belarus	0.221
Germany	0.236	Australia	0.204	Honduras	0.215
Thailand	0.222	Sri Lanka	0.207	Austria	0.225
France	0.228	Romania	0.218	Tajikistan	0.198
UK	0.222	Burkina Faso	0.225	Israel	0.172
Italy	0.184	Niger	0.170	Switzerland	0.202
Myanmar	0.215	Malawi	0.212	Togo	0.240
South Africa	0.195	Kazakhstan	0.205	Somalia	0.162
Tanzania	0.214	Syria	0.162	Denmark	0.202
South Korea	0.222	Mali	0.170	Laos	0.233
Spain	0.207	Chile	0.191	Republic of Serbia	0.223
Colombia	0.233	Netherlands	0.223	Bulgaria	0.213
Kenya	0.186	Ecuador	0.190	Paraguay	0.233
Argentina	0.212	Cambodia	0.231	Papua New Guinea	0.205
Ukraine	0.225	Zambia	0.236	Libya	0.165
Algeria	0.163	Guatemala	0.210	Lebanon	0.153
Uganda	0.205	Senegal	0.211	El Salvador	0.189
Iraq	0.162	Zimbabwe	0.216	Sierra Leone	0.224
Poland	0.228	South Sudan	0.218	United Arab Emirates	0.155
Sudan	0.179	Guinea	0.222	Nicaragua	0.235

offshore $EM < 0.10$. Onshore, the global median EM ($\tilde{EM}$) is 0.210. Among the countries with the highest EM values are Suriname ($\tilde{EM} = 0.255$), Sweden ($\tilde{EM} = 0.250$), Finland ($\tilde{EM} = 0.247$), and Brazil ($\tilde{EM} = 0.238$) (Table 1). In all these countries, the share of the forest area is greater than 50%. In Suriname, which is the country with the world's largest forest area, it is 98.3%.³⁵ In the large

forested areas of Canada and Russia, EM is in a similar range. In contrast, in many island states such as Seychelles and Trinidad and Tobago, $\tilde{EM}$ is below 0.050. Furthermore, low values of EM occur in deserts and steppes. In Oman, Qatar, and the Western Sahara, $\tilde{EM}$ is below 0.15.

These results demonstrate that the surface roughness, which is predominantly influenced by the land use,

drives the spatial pattern of EM on a global scale. Besides, the orography influences EM . In general, at grid cells where the elevation is lower compared to the surrounding grid cells, EM increases. The opposite is true at grid cells where the elevation is higher compared to the neighboring grid cells. Thus, rapid changes of EM occur in mountain areas, where the relative elevation varies greatly. Striking examples of such EM patterns are the Himalaya in Asia and the Rocky Mountains in North America.

3.2 | Temporal variability

The monthly variation of EM is remarkable (Figure 4). The land-sea differences of EM are more pronounced in winter than in summer. Towards the equator, the seasonal EM differences are smaller compared to the high latitudes. Global onshore $\bar{EM}$ ranges from 0.213 DJF (Figure 3A) to 0.200 in JJA (Figure 4C). The high $\bar{EM}$ value in DJF is mainly due to high EM values north of 50° , which frequently exceed 0.25, especially in forested areas.

In many Northern and Central European countries, $\bar{EM}$ is about 0.25. In MAM, EM decreases strongly (Figure 4B). For instance, in Sweden, $\bar{EM}$ is 0.277 and 0.245 in DJF and MAM. This EM trend continues in JJA. It is noticeable that the influence of the relative elevation on EM also decreases in summer as indicated by a lower sd of EM .

South of latitude 30°S , highest EM values occur in JJA. However, the EM values are below the level of the northern hemisphere in DJF. The pattern of EM in SON (Figure 4D) is very similar to the pattern of EM in MAM.

The global $\bar{EM}$ in SON (0.211) is slightly above global $\bar{EM}$ in MAM (0.208). The most important reasons causing the temporal pattern of EM are the atmospheric stratification and wind speed, which both are subject to an annual cycle.^{36,37}

The global sd of EM on an annual scale (sd_a) is displayed in Figure 5A. The results reveal that the inter-annual variability of E is very small. Onshore, median sd_a ($\bar{sd}_a$) is 0.003. In only 16 countries, $\bar{sd}_a > 0.005$. The inter-annual variability of E is only important in some very sparsely populated regions including Alaska, northeastern Russia, and the coast of Greenland. But even there,

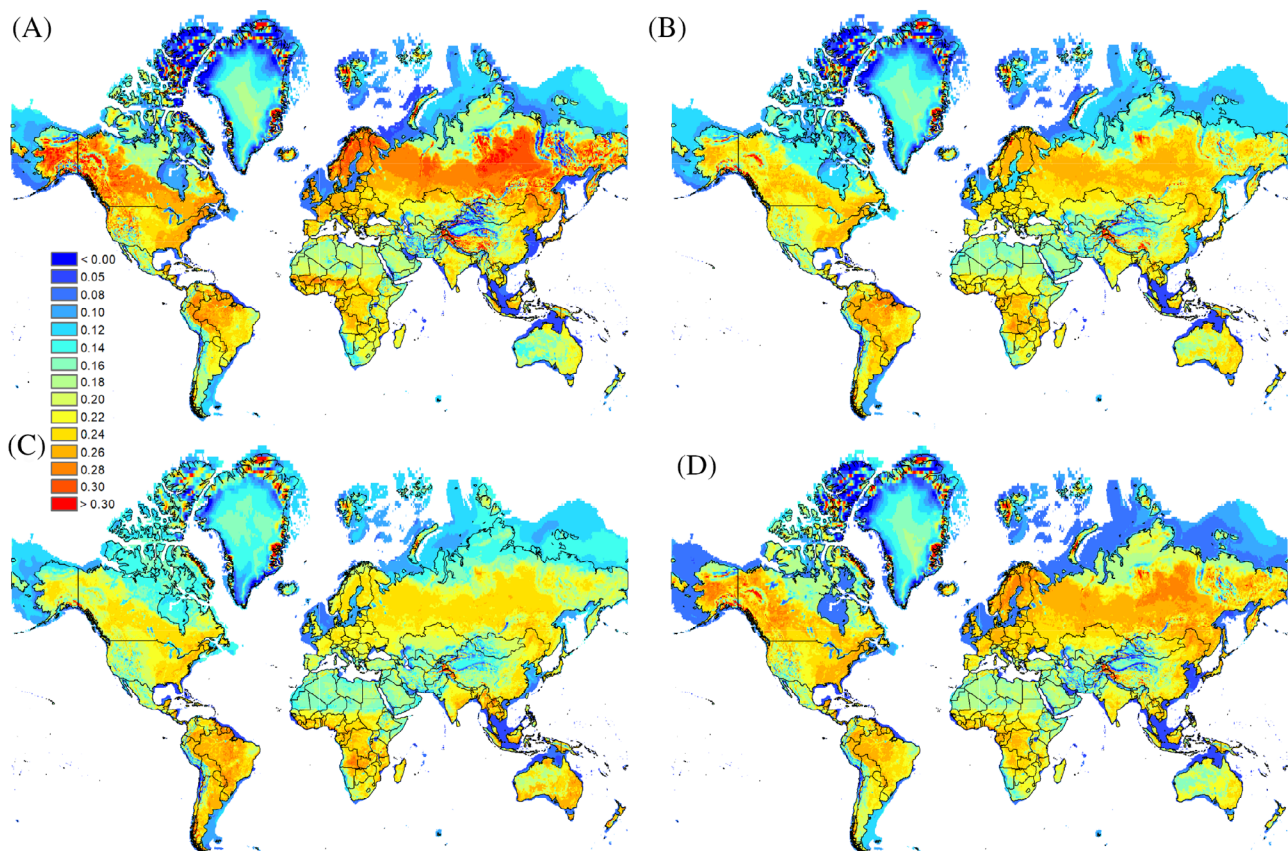


FIGURE 4 Global pattern of the mean power law exponent (EM) in A, winter (EM_{DJF}), B, spring (EM_{MAM}), C, summer (EM_{JJA}), and D, autumn EM_{SON} . The legend values indicate the highest class values [Colour figure can be viewed at wileyonlinelibrary.com]

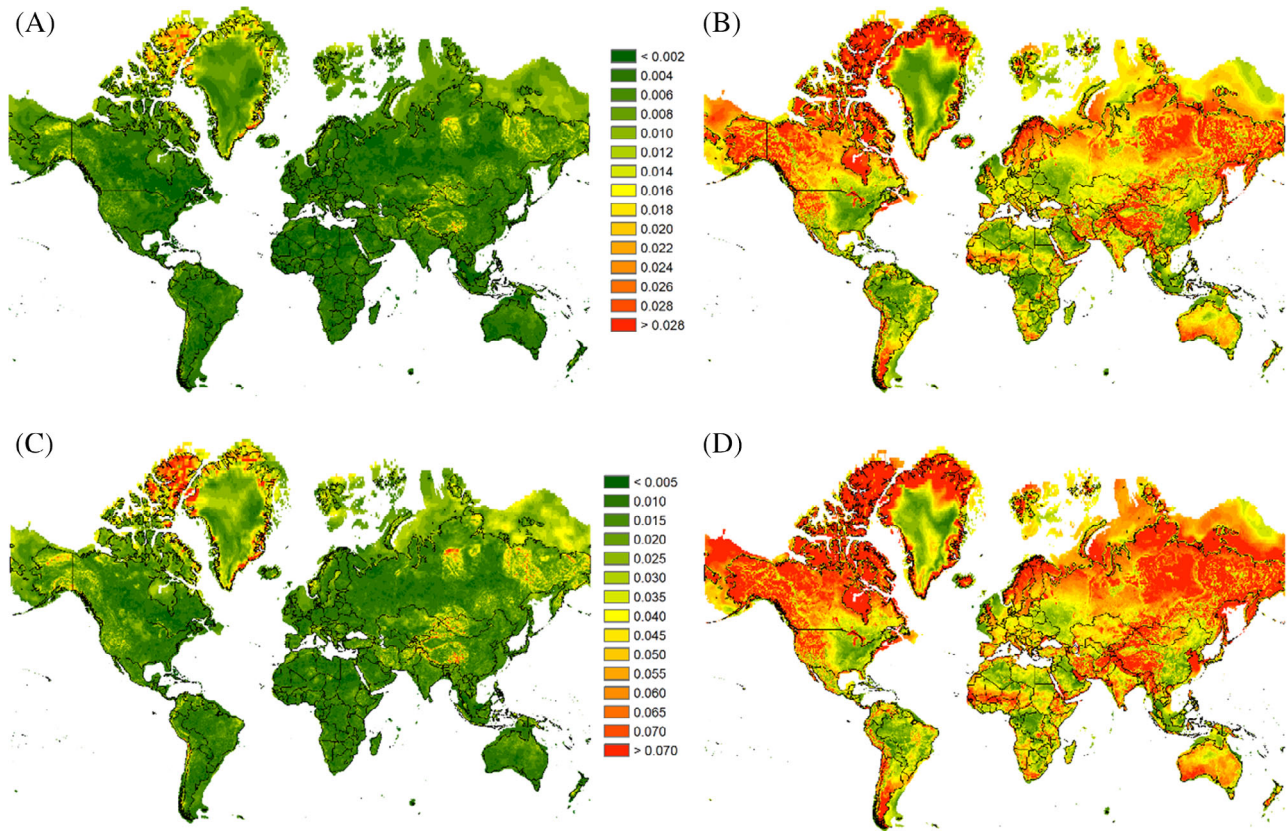


FIGURE 5 Global pattern of the sd of the mean power law exponent (EM) on the A, annual scale (sd_a) and B, monthly scale (sd_m) as well as the range (r) of the mean power law exponent (EM) on the C, annual scale (r_a) and D, monthly scale (r_m). The legend values indicate the highest class values [Colour figure can be viewed at wileyonlinelibrary.com]

the pattern of areas where $sd_a > 0.010$ is patchy. The results reveal that there is no influence of the inter-annual variability of U on E .

In contrast, there is a striking seasonal variability of E in most parts of the world (Figure 5B). The monthly global onshore median of sd ($\tilde{sd}_m$) is 0.015.

The spatial pattern of sd_m is very complex. In 23 countries including Nepal, Sweden, and Canada, $\tilde{sd}_m > 0.020$. These countries can be found at very different latitudes and longitudes. The land-sea distribution does not seem to be a decisive factor for the spatial distribution of sd_m . Whereas $sd_m \approx 0.028$ at the northeast Atlantic coast of the USA, $sd_m < 0.005$ in the northwest Atlantic of Europe.

In Figure 5C, r on the annual scale (r_a) is presented. The global onshore median of r_a ($\tilde{r}_a$) is 0.011. The spatial distribution of r_a confirms that the inter-annual variability of E is often of little importance. However, in mountainous regions including the Andes, the Himalaya, and the Central Siberian plateau, r_a exceeds 0.040 in many cases. $\tilde{r}_a$ values in mountainous countries like Nepal or Andorra usually exceed 0.020. Nevertheless, in the vast majority of countries, $\tilde{r}_a < 0.015$.

As opposed to this, global onshore $\tilde{r}$ on the monthly scale ($\tilde{r}_m$) is much higher being at 0.045 (Figure 5D). In 60 countries, $\tilde{r}_m$ exceeds 0.050. Similar to sd_m , the spatial distribution of r_m is very heterogeneous and independent of the land-sea distribution. The factors determining the spatial r_m distribution are diverse and cannot be unambiguously traced back. North of latitude 60°N and south of latitude 30°S , r_m frequently exceeds 0.050. Similar high r_m values also occur in mountainous regions.

3.3 | Influence on the wind resource

The global onshore median of AER_{WPD} ($\tilde{AER}_{WPD}$) applying EM in the period January 1, 2007 to December 31, 2018 is 8 W/m^2 (Figure 6A). The spatial pattern of ER_{WPD} is very complex, which indicates that it is not possible to define a universally applicable correction factor for the EM -based WPD estimation. The greatest accuracy in the estimation of WPD is achieved near the equator. This is mainly because the seasonal variability of EM in this region is low. In countries such as Papua New

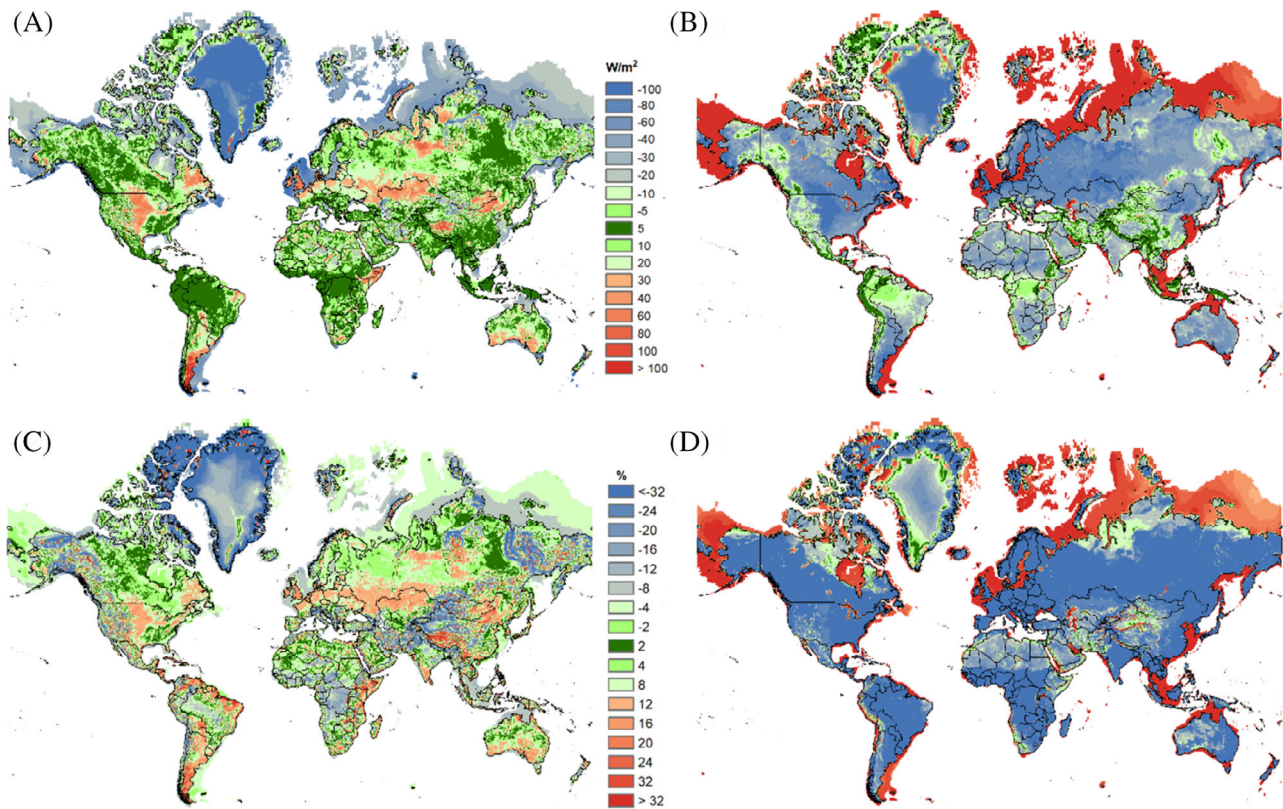


FIGURE 6 Global pattern of the error in the wind power density estimation (ER_{WPD}) using A, the mean power law exponent (EM) and B, assuming $E = 0.14$ ($E014$) as well as the percentage error in the wind power density estimation (PER_{WPD}) using C, the mean power law exponent (EM) and D, assuming $E = 0.14$ ($E014$). The legend values indicate the highest class values [Colour figure can be viewed at wileyonlinelibrary.com]

Guinea, Mexico, and Brazil, AER_{WPD} is often $<5 W/m^2$. With increasing distance from the equator, the pattern of ER_{WPD} becomes patchier. Offshore, there is a tendency towards an underestimation of WPD . In the northwestern European offshore region, which is the most important region for the offshore use of wind energy,³⁸ WPD is often underestimated by more than $100 W/m^2$. In contrast, WPD is often overestimated in onshore regions where the installed wind capacity is high, such as the United Kingdom and the Midwest of the USA. An important reason for the direction and magnitude of ER are the seasonal fluctuations of E .

The global pattern of ER_{WPD} using $E014$ reveals considerable differences between onshore and offshore areas (Figure 6B). Offshore, WPD is greatly overestimated with ER_{WPD} values almost everywhere exceeding $100 W/m^2$. Onshore, WPD is often underestimated. In Northwest Europe and Central and Eastern North America, ER_{WPD} is the lowest ($< -50 W/m^2$). However, there are also some regions where ER is $\approx 0 W/m^2$, indicating good potential for high quality assessments of the wind resource assuming $E014$. This includes large parts of the West Coast of the Americas and regions near the equator.

The reason for the direction and magnitude of ER_{WPD} associated with the use of $E014$ is the bias of $E014$ in the spatial distribution of E .

The spatial pattern of PER when applying EM differs on very small spatial scales (Figure 6C). Especially, in regions with complex orography and rapidly changing land use, areas with positive and negative PER_{WPD} are often close. According to PER_{WPD} , the highest accuracy of the power law is in northeastern Canada, northwestern Russia, and in many deserts around the world. The global onshore median of $APER_{WPD}$ using EM ($\tilde{APER}_{WPD}$) is 7.5%. High accuracy in the estimation of WPD is achieved in large parts of Finland ($\tilde{APER}_{WPD} = 1.1\%$) and desert countries such as Kuwait ($\tilde{APER}_{WPD} = 3.0\%$), Libya ($\tilde{APER}_{WPD} = 3.6\%$), and the Western Sahara ($\tilde{APER}_{WPD} = 4.0\%$). As opposed to that, in countries where the orography is highly diverse including Switzerland ($\tilde{APER}_{WPD} = 15.0\%$), Italy ($\tilde{APER}_{WPD} = 13.8\%$), and Argentina ($\tilde{APER}_{WPD} = 10.3\%$), the accuracy in WPD estimation is lower.

In Figure 6D, PER_{WPD} using $E014$ is presented. Offshore, WPD is frequently overestimated by more than 30.0%. Onshore, WPD is often underestimated by more than 40.0%. The areas where $PER_{WPD} \approx 0.0\%$ are rare. In

desert areas and permafrost regions, the underestimation of WPD is relatively small ($PER_{WPD} \approx -8.0\%$). In contrast, WPD underestimation is very high in the evergreen tropical forests and temperate forests ($PER_{WPD} < -50.0\%$). The global onshore median of $APER_{WPD}$ using $E014$ ($\tilde{A}PER_{WPD}$) is 37.1%. The only country where $\tilde{A}PER_{WPD}$ is quite low is Qatar (1.7%). Elsewhere, $\tilde{A}PER_{WPD}$ exceeds 5.0% anywhere in the world. Substantially inaccurate $\tilde{A}PER_{WPD}$ values occur in Brazil (49.2%) and Congo (56.0%), for example.

3.4 | Influence on the technical potential

The global distribution of ER in the capacity factor estimation (ER_{cf}) using EM of the period January 1, 2007 to December 31, 2018 is displayed in Figure 7A. Offshore ER_{cf} is often in the range around 0.000. The largest contiguous onshore area where $ER_{cf} \approx 0.000$ can be found in Northern Europe. In northwestern parts of North America and near the equator, the use of EM results in only minor errors. A slight underestimation of cf ($ER_{cf} < -0.020$) occurs in parts of North Africa, northern China, and Mongolia. Areas, where ER_{cf} is positive, are less abundant than

areas where ER_{cf} is negative. Onshore, the global median of ER_{cf} using EM ($\tilde{E}R_{cf}$) is -0.009 . In North African and Arab countries such as Kuwait ($\tilde{E}R_{cf} = -0.038$), Niger ($\tilde{E}R_{cf} = -0.034$), and Saudi-Arabia ($\tilde{E}R_{cf} = -0.025$) cf is underestimated. A moderate cf overestimation was found for Cuba ($\tilde{E}R_{cf} = 0.013$) and Kenya ($\tilde{E}R_{cf} = 0.011$).

The pattern of ER_{cf} using $E014$ (Figure 7B) is similar to ER_{WPD} . Offshore, cf is overestimated by more than 0.050 and onshore ER_{cf} is mostly negative. In Northwest Europe and eastern North America, $ER_{cf} \approx -0.010$. A reasonable quality of wind resource assessment assuming $E014$ can be achieved in large parts of the West Coast of America and regions near the equator where $ER_{cf} \approx 0.000$. The global onshore $\tilde{E}R_{cf}$ is -0.061 . In many island states, cf is drastically overestimated including New Zealand ($\tilde{E}R_{cf} = 0.168$) and Trinidad and Tobago ($\tilde{E}R_{cf} = 0.136$). In contrast, in many European countries, including Sweden ($\tilde{E}R_{cf} = -0.144$) and Germany ($\tilde{E}R_{cf} = -0.123$), cf is considerably underestimated.

The spatial distribution of PER_{cf} using EM is presented in Figure 7C. There are numerous areas where PER_{cf} is in the range $\pm 2\%$ including the northern part of Europe and the northeastern part of North America. The accuracy of the power law is also high in many offshore

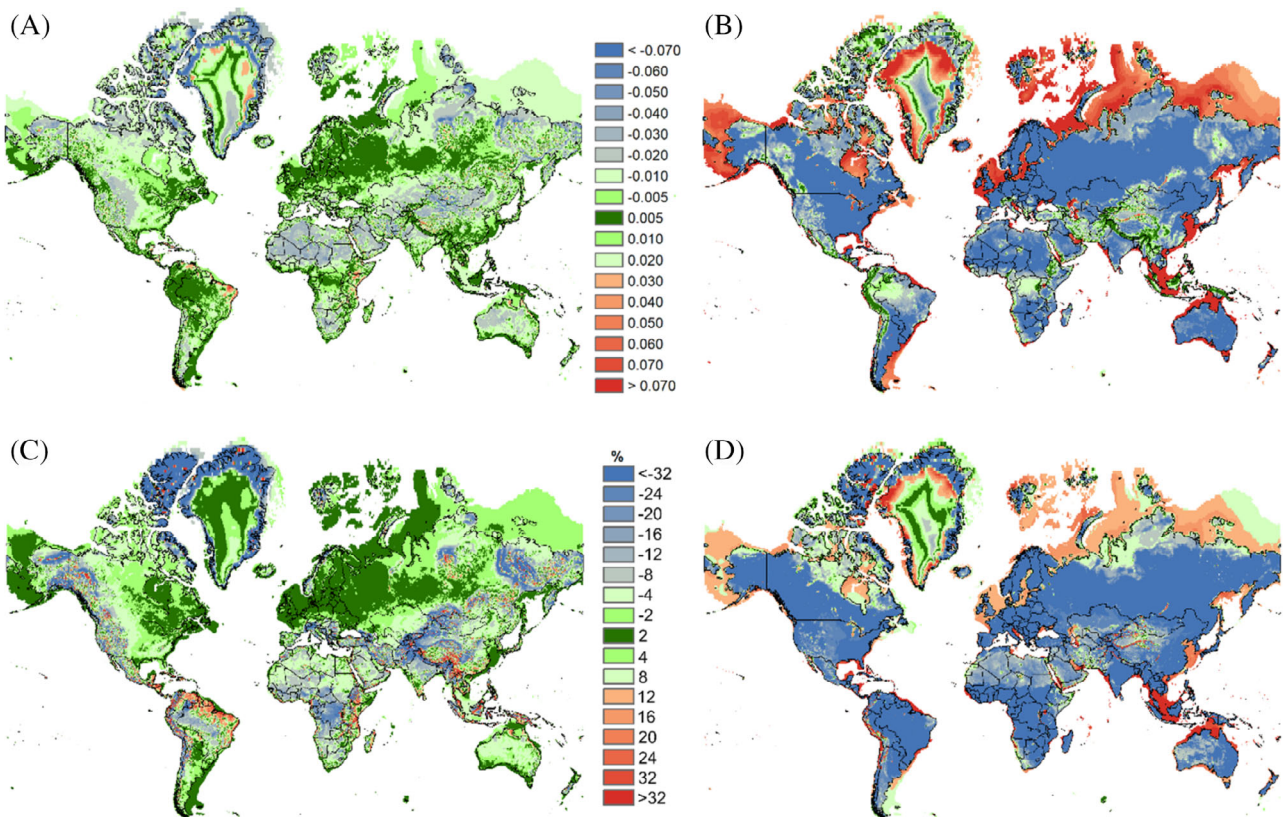


FIGURE 7 Global pattern of the error in the capacity factor estimation (ER_{cf}) using A, the mean power law exponent (EM) and B, assuming $E = 0.14$ ($E014$) as well as the percentage error in the capacity factor estimation (PER_{cf}) using C, the mean power law exponent (EM) and D, assuming $E = 0.14$ ($E014$). The legend values indicate the highest class values [Colour figure can be viewed at wileyonlinelibrary.com]

regions. Towards the equator, $APER_{cf}$ increases. Areas, where cf is underestimated, are often close to areas where cf is overestimated. This indicates that the orography is an important factor driving the spatial distribution of PER_{cf} . The global onshore $\tilde{APER}_{cf}$ is 5.5%.

Applying $E014$, onshore PER_{cf} is frequently below -16.0% (Figure 7D). Slightly higher PER_{cf} values usually occur only in deserts and steppes. The global onshore $\tilde{APER}_{cf}$ is 36.9%, which indicates a much greater inaccuracy of the power law due to the use of $E014$ compared to EM .

Supporting information on the country-related cf errors by applying the power law using EM and $E014$ are presented in Figure S1.

4 | CONCLUSIONS

This study provides a global overview of the errors in wind energy assessments resulting from applying mean and constant power law exponents. It was found that, onshore, the global median absolute percentage error in the wind turbine capacity factor estimations are 5.5% applying mean power law exponents and 36.9% assuming the power law exponent being 0.14.

The application of mean and constant power law exponents is not suitable to reproduce the spatiotemporal variability of daily power law exponents. It was revealed that the errors produced from the use of mean and constant power law exponents differ on the spatial scale. In many regions, where the wind resource is abundant, using only mean exponents leads to minor errors in estimating the capacity factor. In these regions, a reasonable assessment of the technical wind potential using mean exponents is possible, even in the absence of long-term wind speed measurements. The prerequisite for mean value-based assessments is the availability of spatially explicit estimates of mean power law exponents available from global wind speed models such as GloWiSMo.¹² GloWiSMo contains a global high spatial resolution (horizontal resolution: $250\text{ m} \times 250\text{ m}$) grid of mean power law exponents. Using values of mean power law exponents from GloWiSMo enables minimizing the error caused by the exponent's spatial variability.

At the local scale, there are further influencing factors driving the spatiotemporal pattern of the power law exponent that are not integrated at the spatial resolution of the ERA5 data. Locally, the errors in assessing wind power densities and capacity factors may deviate from this study's error values.

Future investigations could integrate small-scale surface properties into the error assessment of the current study. This may offer the potential to derive an atlas of spatially explicit correction factors for using mean power law coefficients in wind energy assessments.

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SYMBOLS

$\tilde{AER}$	median of absolute error
$\tilde{APER}$	median of absolute percentage error (%)
$\tilde{EM}$	median of mean power law exponent
$\tilde{ER}$	median of error
$\tilde{r}$	median of range
$\tilde{sd}$	median of standard deviation
AD	air density (1.225 kg/m^3)
AER	absolute error
$APER$	absolute percentage error (%)
cf	capacity factor
E	power law exponent
$E014$	power law exponent = 0.14
EM	mean power law exponent
ER	error
EV	variant of power law exponent $\{E, EM, E014\}$
$f()$	probability density function
$F()$	cumulative distribution function
$F^{-1}()$	quantile function
h	height (m)
PER	percentage error (%)
PR	rated power (MW)
$PW(U)$	wind turbine power curve (MW)
PWM	mean wind turbine power output (MW)
r	range
sd	standard deviation
t	Time (h)
u	zonal wind vector component (m/s)
U	wind speed (m/s)
v	meridional wind vector component (m/s)
WPD	wind power density (W/m^2)
WPV	wind potential variable $\{cf, WPD\}$
x	x direction
y	y direction
α	first Wakeby scale parameter
β	first Wakeby shape parameter
γ	second Wakeby scale parameter
δ	second Wakeby shape parameter
ε	Wakeby location parameter

SUBSCRIPTS

a	annual scale
E	time-dependent power law exponent
g	near-surface (10 m)
he	height above ground level

<i>hub</i>	wind turbine hub ($\approx$ 80–140 m)
<i>m</i>	monthly scale
<i>WPD</i>	wind power density
<i>10</i>	10 m height above ground level
<i>100</i>	100 m height above ground level

CONFLICT OF INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author, upon reasonable request.

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of this article.

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